

Predicting Hawaii Wildfire Risk



Wildfires are an escalating challenge in Hawaii, threatening communities, cultural sites, infrastructure, and native ecosystems. On average 17,000 acres burn each year — around 0.4% of state land area. Climate change is intensifying this risk through various factors including longer droughts, extended heat periods that prevent evening humidity recovery, and the spread of invasive vegetation types that fuel faster fire growth.

Hawaii's unique combination of diverse microclimates, topographies, and land uses makes anticipating and managing wildfires an especially difficult challenge, unsuitable for many of the existing prediction models designed for the continental United States. A model tailored specifically to Hawaii could improve fire management by identifying high-risk conditions and guiding resource allocation.

In this first stage of this project, we compiled and cleaned information on past fire occurrence, climate and seasonal patterns, topography, and vegetation. In combining these into a single dataset and testing a baseline XGBoost model, the foundations have been laid for future development of a predictive system that may eventually highlight high-risk areas with high spatial detail.

Data Preparation

Cleaning and combining together data from multiple domains has defined the bulk of the work so far. The main sources included:

- **Wildfire boundaries:** Polygons of fire perimeters across the main islands (1999–2022).
- **Topography:** 30m rasters of slope and elevation.
- **Climate teleconnections:** Monthly Oceanic Niño Index (ONI) and sea-surface temperature anomalies.
- **Tourism:** Monthly domestic and international arrivals by island, plus statewide totals.
- **Vegetation:** 30m rasters of terrestrial ecological systems from 2002 and 2014.
- **Rainfall:** 250m monthly rasters (processed but not yet integrated).

The foundation of the modelling dataset was a downsampled spatial grid derived from vegetation raster data. Vegetation rasters from 2002 and 2014 were aligned to a common coordinate reference system and downsampled by a factor of 20 to reduce computational complexity while preserving dominant land cover types. Vegetation type classifications for 1999–2008 were assigned based on the 2002 vegetation dataset, while the 2014 dataset was used for classifications between 2009–2022. Each downsampled pixel was converted to a polygon grid cell, creating spatial units for analysis.

Static environmental characteristics were assigned to each grid cell through spatial overlay operations. Elevation and slope summary statistics (mean, median, minimum, maximum, and standard deviation) were calculated for each polygon by extracting values from the aligned rasters.

The climate teleconnection indices and tourism arrival data were merged at a monthly resolution. Teleconnection variables were lagged by 1, 3, and 6 months to capture delayed climate effects on fire conditions. Monthly dummy variables were created to account for seasonal patterns in fire occurrence.

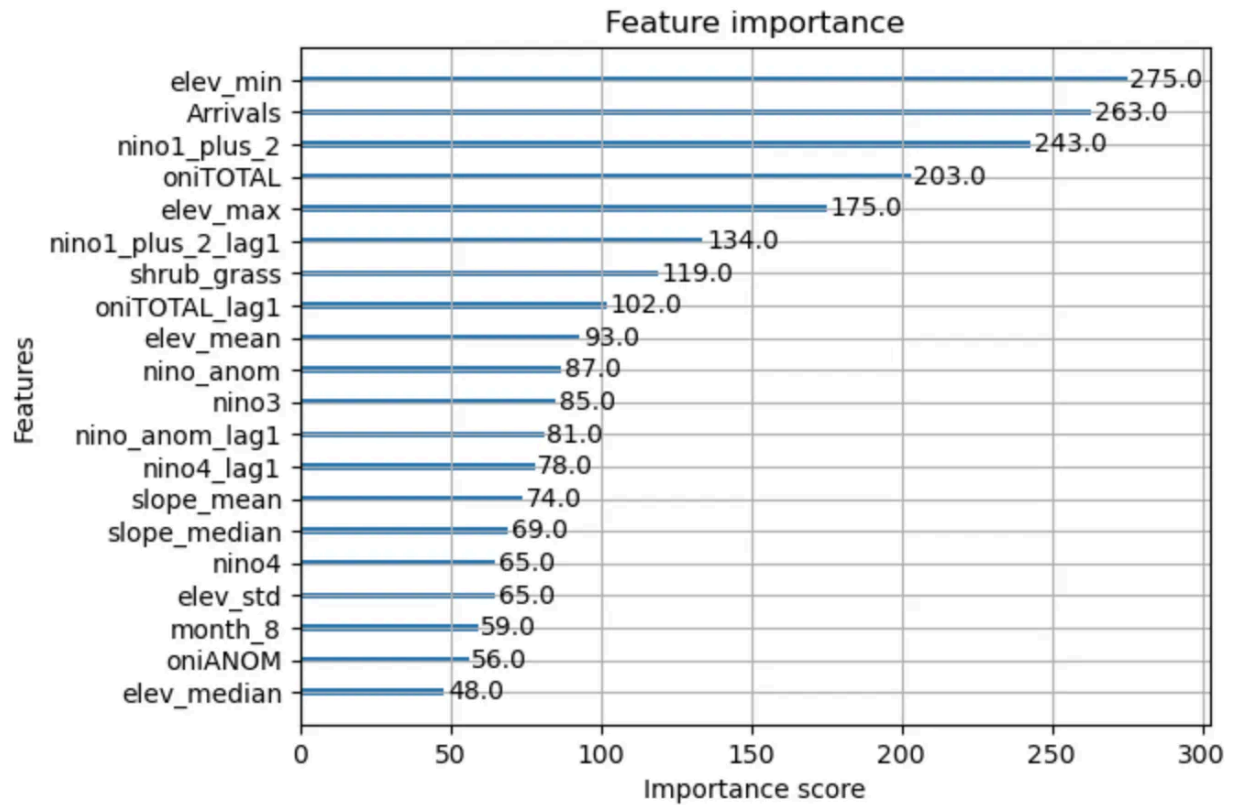
Wildfire occurrence was determined through spatial-temporal joins between the grid cells and historical fire polygons. Fire events were matched to grid cells based on spatial intersection and temporal alignment (same year and month). For cells with multiple overlapping fire events in the same time period, duplicate records were removed to create binary fire occurrence indicators.

The final integrated dataset contained over 2 million records representing unique combinations of grid cells, years (1999–2022), and months. Each record included environmental variables (elevation, slope, vegetation type), temporal climate variables (lagged teleconnection indices), anthropogenic factors (tourism data), and the binary fire occurrence target variable. To address computational constraints and class imbalance, the dataset was reduced by randomly sampling non-fire observations while retaining all fire events.

Baseline Model

We selected XGBoost for the initial modelling approach due to its effectiveness with heterogeneous datasets containing mixed data types, its ability to handle class imbalance through built-in weighting mechanisms, and its robustness to the spatial and temporal correlations inherent in environmental data. XGBoost is also particularly well-suited for capturing any complex non-linear relationships between environmental variables and fire occurrence.

Temporal cross-validation was used in the modelling process to reflect real-world forecasting scenarios, where predictions of future fire risk would be made based on past observations. Training data included years 1999–2020, while 2021–2022 were held out for final testing. Hyper-parameter tuning was conducted using a time series split approach with parameters including maximum tree depth and learning rate optimised for precision-recall performance given the class imbalance.



Future Development

This baseline model represents the first stage of a project aimed at developing and deploying a machine learning model for the Hawaii Wildfire Management Organization (HWMO) to predict wildfire risk at high spatial resolution. Several avenues exist for improvement and enhancement: incorporating rainfall data and potentially drought indices (if sufficient coverage exists for the Hawaiian Islands); including additional environmental data sources such as humidity, and historical fire frequency/intensity; using island-specific tourism data rather than statewide aggregates; additional feature engineering to include spatially relevant predictors (e.g., distance to roads or rivers, characteristics of neighbouring polygons); experimentation with spatially aware models or ensemble methods; and eventually developing real-time prediction capabilities to support dynamic risk assessments. In the next stage, outputs will be integrated into decision support tools, producing maps of wildfire risk and identifying key vulnerable locations to inform prioritisation of interventions and resources.